Budget Constraint

Reference: Microeconomics by Mankiw (8e)

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The Basic Idea

- A consumer can choose from a bundle of goods $I \in \{1, ...N\}$.
- However, the have to allocate their budget in terms of how many of each good they want to consumer, thus generating a bundle $X \in \{x_1, ..., x_N\}$.
- Each good has a price, the set of which is given by $P \in \{p_1, ...p_N\}$.
- Thus, the total expenditure is given by $\sum_{i=1}^{N} p_i x_i$.

Why budget constraints?

- Off the bat, we can see that if there were no constraints on budget, one should consume infinite (or extremely large number) of each good.
- This makes our analysis both unrealistic and uninteresting.
- Now suppose a person has a budget m.

How it looks

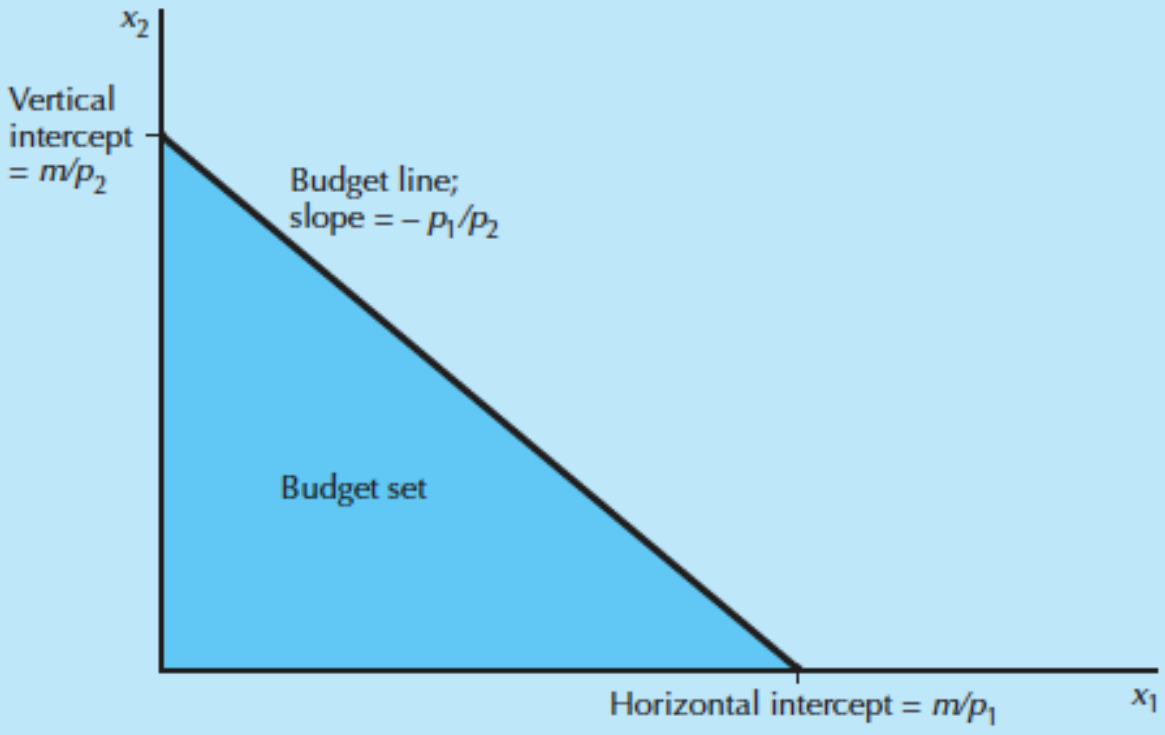
$$\sum_{i=1}^{N} p_i x_i \leq m$$

- Suppose, $N = 2$. This is enough for a lot of analysis.

$$p_1 x_1 + p_2 x_2 \leq m$$

- Sometimes x_2 is thought of as a **composite good**.

How it looks



The budget set. The budget set consists of all bundles that are affordable at the given prices and income.

If we want to represent it a little differently?

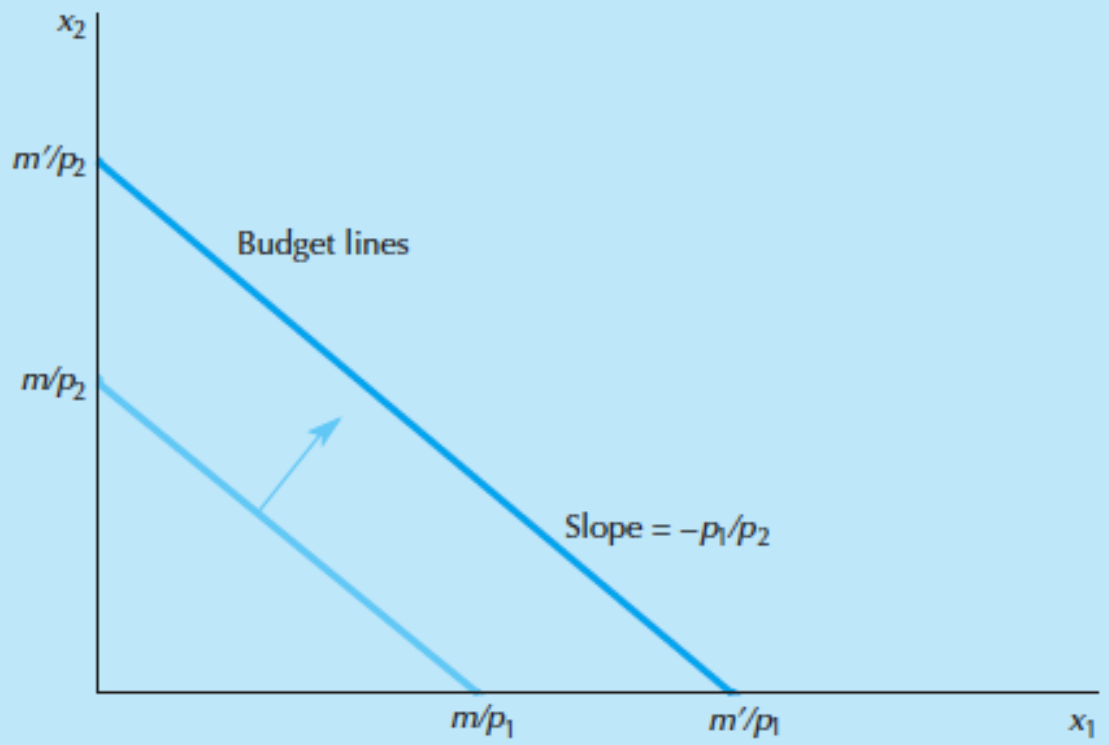
$$x_2 = \frac{m}{p_2} - \frac{p_1}{p_2} x_1$$

- Does p_i have to be less than m?

What about a change in consumption?

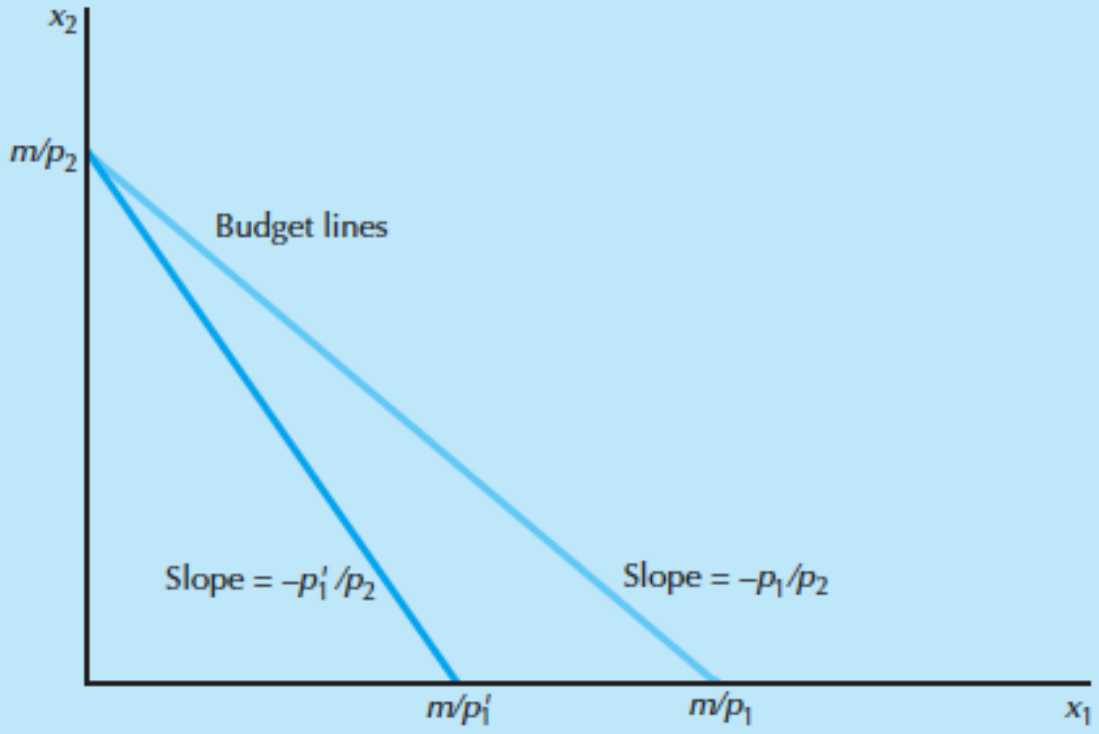
- $p_1 \Delta x_1 + p_2 \Delta x_2 = 0$. Why?
- $\frac{\Delta x_2}{\Delta x_1} = -\frac{p_1}{p_2}$
- This is a cleaner representation of opportunity cost.

Changes in Budget



Increasing income. An increase in income causes a parallel shift outward of the budget line.

Changes in Price

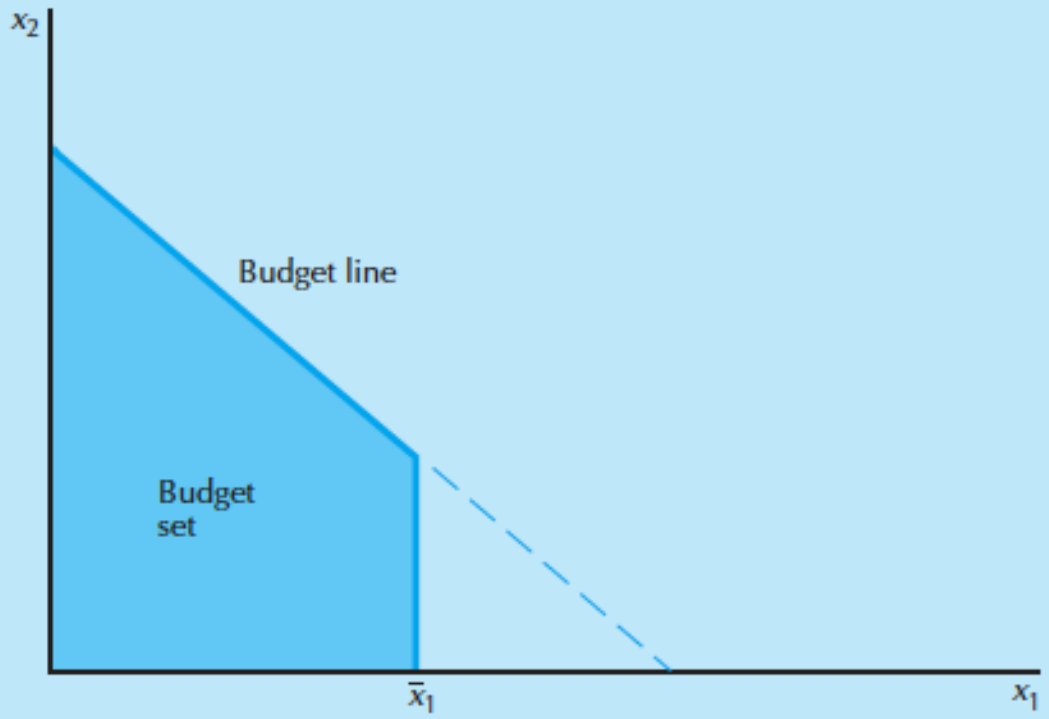


Increasing price. If good 1 becomes more expensive, the budget line becomes steeper.

The Numeraire

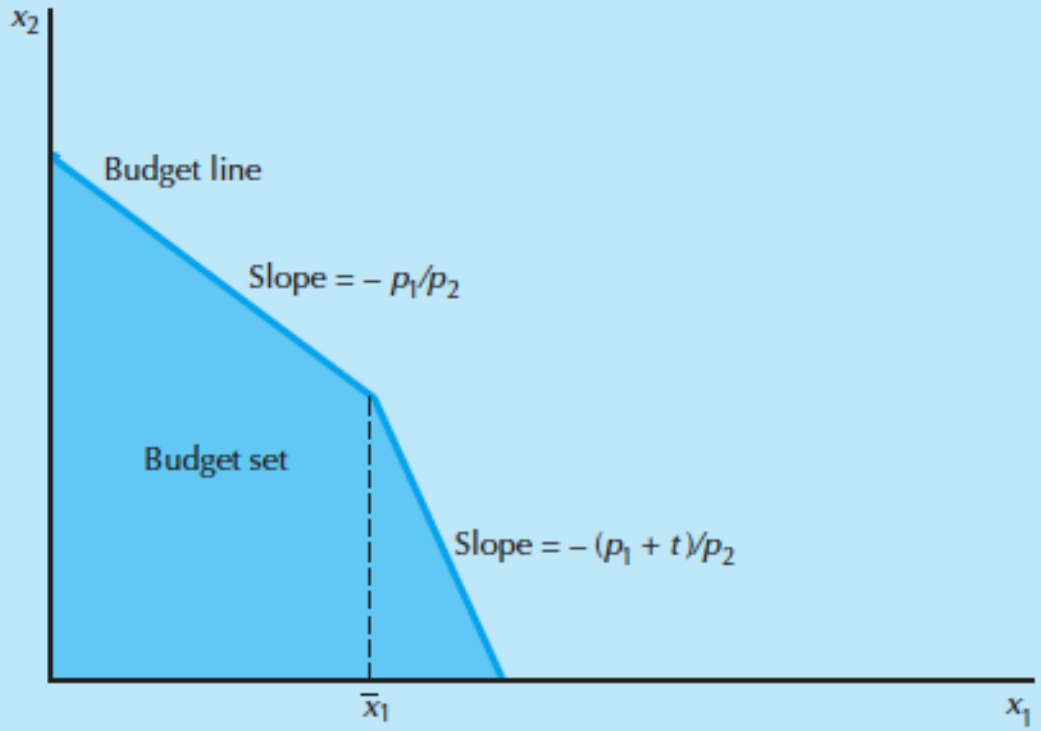
- Sometimes we peg all prices to the price of one good. This gives us something to compare other things to.
- $\frac{p_1}{p_2}x_1 + x_2 = \frac{m}{p_2}$.
- Sometimes, one can also write it as $\frac{p_1}{m}x_1 + \frac{p_2}{m}x_2 = 1$

Rationing



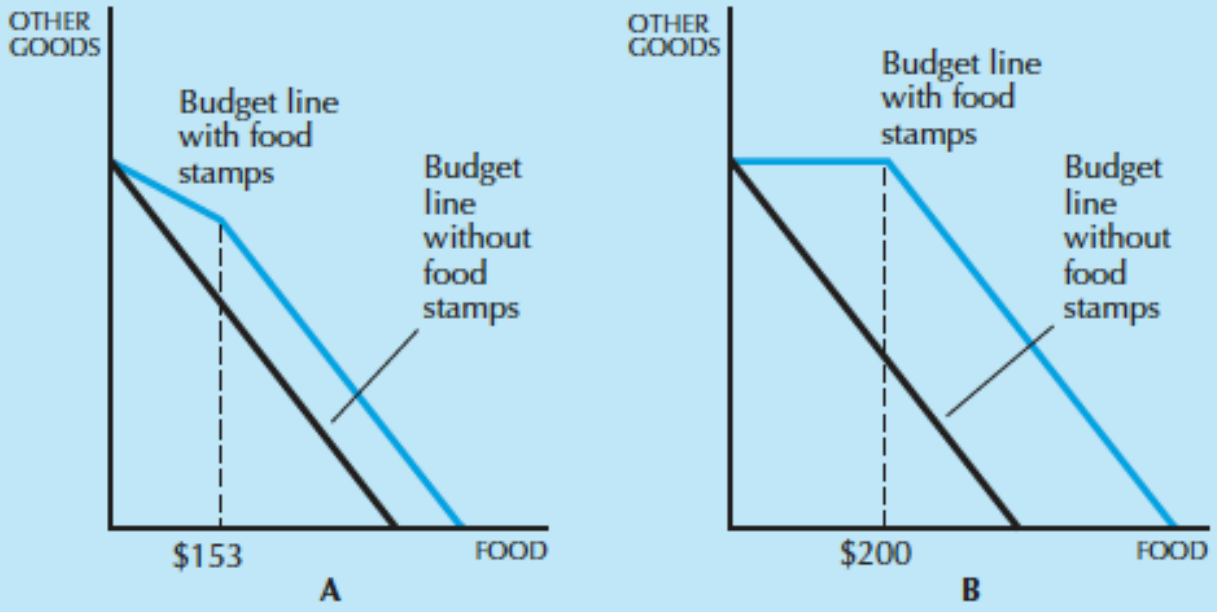
Budget set with rationing. If good 1 is rationed, the section of the budget set beyond the rationed quantity will be lopped off.

Taxes



Taxing consumption greater than $\overline{x}_1$. In this budget set the consumer must pay a tax only on the consumption of good 1 that is in excess of $\overline{x}_1$, so the budget line becomes steeper to the right of $\overline{x}_1$.

Food Stamps



Food stamps. How the budget line is affected by the Food Stamp program. Part A shows the pre-1979 program and part B the post-1979 program.